Tutorial Questions 03 – Functions

Dr Killian O'Brien

6G4Z3006 Calculus

Semester 2 — 2025/26

When possible you should work on the questions in advance and be prepared for discussions on them with your colleagues and tutor. Some questions are straightforward calculation while others require more discussion and thought. The computer icon indicates where computer calculation, plotting or programming may be of use.

Further questions can be found in the lecture notes as well as the recommended unit resources (see Moodle).

Domains and co-domains

1. Consider a function $\phi : X \to Y$, where X, Y are subsets of $\mathbb{R}$, defined by the formula

$$\phi(x) = \frac{x^2+1}{2x-3}.$$

   Can you choose the subsets X and Y so that the function ϕ is variously

   a) injective and surjective, (i.e. bijective)
   b) surjective but not injective,
   c) injective but not surjective,
   d) neither injective nor surjective.

   If X and Y are both the empty set, i.e. $X = Y = \{\}$, is ϕ injective and/or surjective?

2. For a function $\psi : X \to Y$ defined by $\psi(x) = \tan(x)$ answer parts (a) - (d) of question 1.

3. Any $n \times m$ matrix A will determine a function (called a linear transformation) $T_A : \mathbb{R}^m \to \mathbb{R}^n$ defined by

$$T_A(\mathbf{x}) = A\mathbf{x},$$

   where $A\mathbf{x}$ denotes the left-multiplication of the column vector $\mathbf{x}$ by the matrix A, and $\mathbb{R}^m$ denotes the set (actually a vector space) of all m-component column vectors, and likewise for $\mathbb{R}^n$. You will learn all about such linear transformations later in your Linear Algebra lectures. For now, for some small values of n and m, can you exhibit matrices A such that the linear transformations T_A will be

   a) injective and surjective, (i.e. bijective)
   b) surjective but not injective,
   c) injective but not surjective,
   d) neither injective nor surjective.

Composition of functions and inverses

4. Consider the functions f, g, h, all $\mathbb{R} \to \mathbb{R}$, defined by the formulas

$$f(x) = -2x-3, \quad g(x) = x^2+3x+1, \quad h(x) = x/2.$$

   Find the formulas, simplified as far as possible, that represent the functions $f \circ g \circ h$, $g \circ f \circ h$, $h \circ g \circ f$, $f \circ f$, $g \circ g$ and $h \circ h$.

5. Can you compose your own relatively simple, but not trivial, examples to show that composition of functions is not commutative, i.e. that in general $f \circ g \neq g \circ f$?

6. Consider the functions $f : (0, \infty) \to (0, \infty)$ and $g : (0, \infty) \to \mathbb{R}$ defined by

$$f(x) = x^2, \quad g(x) = \log(x).$$

Find the definitions of the functions $g \circ f$ and $(g \circ f)^{-1}$ and confirm in this case that $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.

Partial fractions

7. Obtain the partial fraction expansions of the following rational function expressions

a)

$$-\frac{13\,t - 23}{2\,(3\,t - 1)(t - 3)},$$

b)

$$-\frac{9\,t^2 + 10\,t + 15}{3\,(t^2 + 1)(3\,t - 5)},$$

c)

$$\frac{2\,t^2 - 11\,t + 14}{(5\,t - 4)(t + 1)^2},$$

d)

$$-\frac{25\,t + 14}{(2\,t^2 + 5\,t + 5)(t - 4)}.$$